

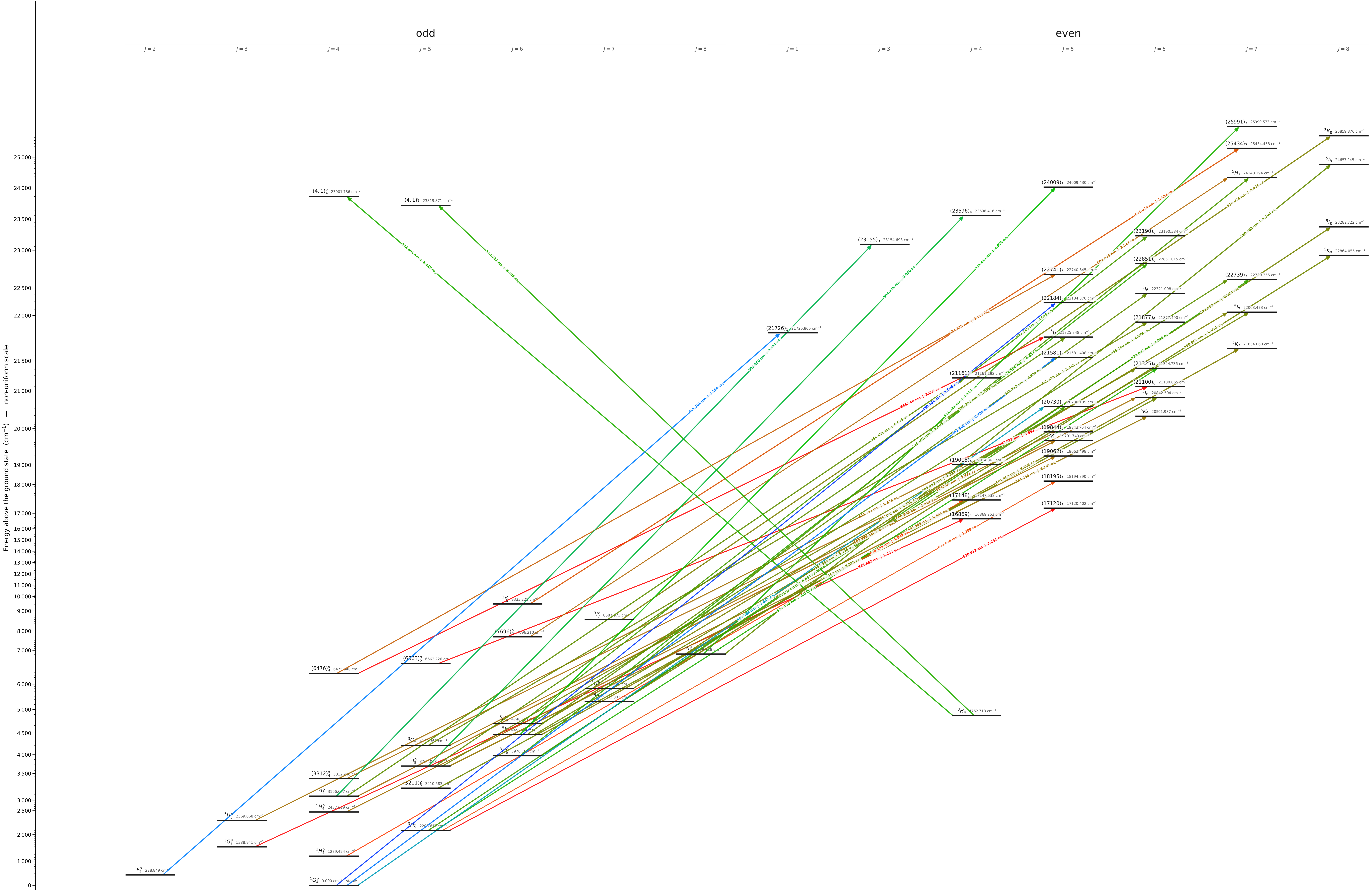
Cerium

Ce (Ce I). Strongest classified transitions below 2 μm (NIST): 62 levels, 45 lines.

Ce ⁺ ionisation limit 44672.00 cm⁻¹ = 5.53862 eV (223.854 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle U||er||U\rangle$ in ea₀.
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle U er U\rangle$ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
³ f ₇ - (23155) ₃	501.0501	500.9103	598.3284	5.191	6.200e+07	-	-	-
³ f ₇ - (23596) ₄	504.2252	504.0846	594.5606	5.000	4.390e+07	-	-	-
³ f ₇ - (24009) ₅	511.4128	511.2704	586.2044	4.976	3.410e+07	-	-	-
³ f ₇ - (25991) ₇	521.3372	521.1920	575.0453	7.111	4.820e+07	-	-	-
³ H ₄ - (4, 1) ₂ ^o	522.4915	522.3460	573.7748	6.417	6.500e+07	-	-	-
³ H ₄ - (4, 1) ₃ ^o	524.7374	524.5913	571.3191	6.206	4.910e+07	-	-	-
³ H ₅ - ³ H ₇	545.0754	544.9239	550.0018	6.403	3.420e+07	-	-	-
³ f ₇ - (21161) ₄	556.6508	556.4962	538.5647	5.625	4.130e+07	-	-	-
³ f ₇ - ³ f ₅	556.7513	556.5968	538.4674	5.076	2.750e+07	-	-	-
³ f ₇ - ³ f ₆	560.2832	560.1277	535.0731	9.794	6.500e+07	-	-	-
(3211) ₂ ^o - ³ f ₆	567.1532	566.9958	528.5917	6.373	3.470e+07	-	-	-
³ f ₇ - ³ K ₈	569.8573	569.6993	526.0834	8.934	5.140e+07	-	-	-
³ H ₅ - ³ f ₆	572.0623	571.9036	524.0556	8.924	5.070e+07	-	-	-
³ f ₇ - ³ K ₉	578.9750	578.8145	517.7986	8.426	4.360e+07	-	-	-
³ f ₇ - ³ K ₇	581.4527	581.2915	515.5922	6.608	3.000e+07	-	-	-
³ f ₇ - ³ K ₆	594.2502	594.0856	504.4886	6.107	2.770e+07	-	-	-

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

Γ/2π: natural linewidth of the upper level, 1/(2πτ₁).